

Answer Key — Biology Grade 10

Sub-Strand 1.2: Chemicals of Life • Lesson 5: Water, Mineral Salts and Enzymes in Digestion

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	D	B	D	A	B	A

USE AFTER: PREDICT PHASE — Predict: Wanjiru's blood glucose graph and prediction about starch and saliva

Q1. Wanjiru's blood glucose rose from 4.5 to 8.0 mmol/L after she ate ugali, which is mostly starch. Why must starch be digested before it can raise her blood glucose?

- A. Starch is a very large insoluble molecule and must be broken down into small soluble sugars first**
- B. Starch is already glucose, but it needs to be warmed to 37 degrees C first
- C. Starch turns directly into blood once it is swallowed
- D. Saliva only makes the food wet so it can be swallowed, and this is all that is needed

Correct answer: A. Digestion breaks large insoluble food molecules into small soluble ones. Starch is too large to enter the blood, so it must first become sugars such as glucose.

USE AFTER: OBSERVE PHASE — Observe: saliva and starch investigation, control tube B (starch + boiled saliva or water)

Q2. In an investigation, tube A has starch and saliva. Tube B has starch and boiled saliva. Both are kept at about 37 degrees C. Why is tube B included?

- A. To make sure there is more liquid to test with iodine
- B. To show that boiled saliva digests starch faster than fresh saliva
- C. To test whether Benedict's solution can change colour without heating
- D. To show that the changes in tube A are caused by active saliva and not by warmth or time alone**

Correct answer: D. The control differs only in having boiled saliva. If tube B stays unchanged while tube A changes, the change must be due to something active in saliva (the enzyme), not warmth or time.

USE AFTER: OBSERVE PHASE — Observe: Steps 5 and 6, iodine tests at 5 and 10 min and Benedict's test at 10 min

Q3. At 10 minutes, a sample from the tube of starch and fresh saliva at 37 degrees C gives a yellow-brown colour with iodine and a brick-red colour with Benedict's solution after heating. What do these results show?

- A. Starch is still present, and the saliva has turned into a reducing sugar
- B. Starch has been used up and a reducing sugar is now present**
- C. Starch is still present and no sugar has formed
- D. Starch has increased and Benedict's solution has turned it into protein

Correct answer: B. Iodine stays blue-black while starch is present, so yellow-brown means starch has gone. A brick-red Benedict's result means reducing sugar has formed from the starch.

USE AFTER: EXPLAIN PHASE — Explain: claim-evidence-reasoning for how saliva changed the starch, and why boiled saliva did not work

Q4. A tube of starch and saliva was boiled before the test and stayed blue-black with iodine after 10 minutes. What is the correct explanation?

- A. Boiling turned the enzyme into starch
- B. Boiling made the starch molecules too big to be digested
- C. Boiling used up all the enzyme by making it work too fast
- D. Boiling denatured the enzyme, so it could no longer digest starch**

Correct answer: D. Salivary amylase is a protein. Boiling changes its shape (denatures it), so it cannot digest the starch and the iodine stays blue-black.

USE AFTER: EXPLAIN PHASE — Explain: discussion (b) on why the enzyme did not work at 0 degrees C or after boiling

Q5. A tube of starch and fresh saliva is placed in ice at 0 degrees C. After 10 minutes it is still blue-black with iodine. What does this tell us about the enzyme?

- A. The enzyme is inactive at this low temperature**
- B. The enzyme works fastest at 0 degrees C
- C. The enzyme is not needed to digest starch
- D. The enzyme has been denatured and destroyed for ever by the cold

Correct answer: A. At 0 degrees C the enzyme is inactive, so the starch is not digested. Enzymes work best at about 37 degrees C in the body, and boiling is what denatures them.

USE AFTER: EXPLAIN PHASE — Explain: discussion (c) on the roles of water and mineral salts

Q6. Water is an inorganic compound found in the starch solution and in saliva. What is its main role in the digestion of starch?

- A. It is the enzyme that breaks starch into sugar
- B. It is the medium in which the enzyme and starch react and in which the digested sugars are carried**
- C. It makes the enzyme a protein
- D. It changes the iodine from blue-black to yellow-brown

Correct answer: B. Water is the medium in which reactions happen and it carries the digested sugars. It is not the enzyme and does not act as the iodine indicator.

USE AFTER: MODEL BUILDING PHASE — Model: revising the class flow-diagram, 'Digestion by enzymes' box

Q7. A group is revising the 'Digestion by enzymes' box in the class model. Which statement about salivary amylase should they write?

- A. It is a protein catalyst that speeds up the digestion of starch to maltose, and it is not used up**
- B. It is a lipid that stores energy in the mouth
- C. It is a carbohydrate that is used up as it digests starch
- D. It is a mineral salt that works best at 0 degrees C

Correct answer: A. Amylase is an enzyme, which is a protein that acts as a biological catalyst. A catalyst speeds up a reaction and is not used up. It digests starch to maltose, which is then broken to glucose.